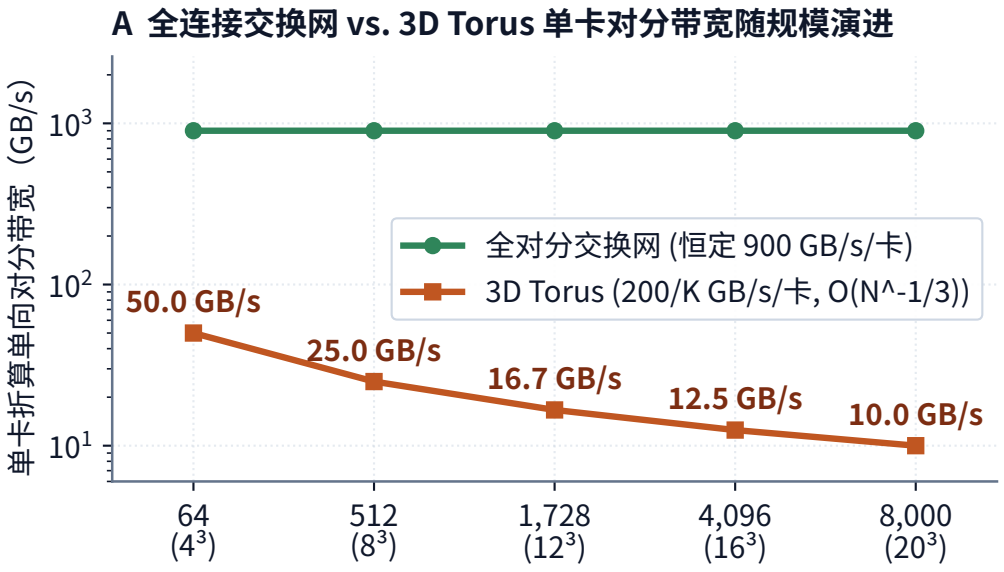
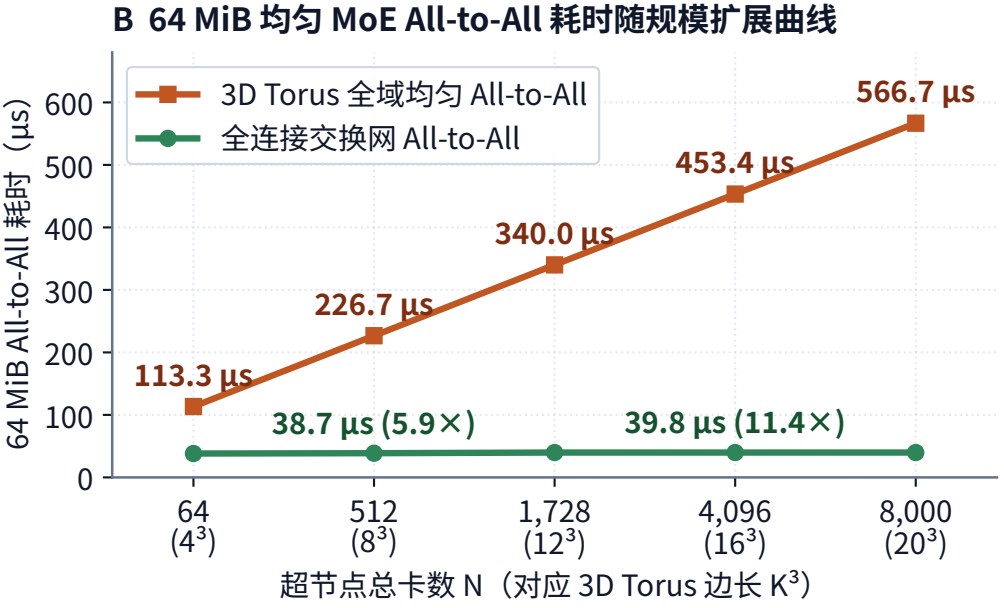

A 全连接交换网 vs. 3D Torus 单卡对分带宽随规模演进
单卡折算单向对分带宽（GB/s）
10^3
10^2
10^1
全对分交换网 (恒定 900 GB/s/卡)
3D Torus (200/K GB/s/卡, O(N^-1/3))
50.0 GB/s
25.0 GB/s
16.7 GB/s
12.5 GB/s
10.0 GB/s
64 (4^3)
512 (8^3)
1,728 (12^3)
4,096 (16^3)
8,000 (20^3)


B 64 MiB 均匀 MoE All-to-All 耗时随规模扩展曲线
64 MiB All-to-All 耗时（μs）
0
100
200
300
400
500
600
3D Torus 全域均匀 All-to-All
全连接交换网 All-to-All
113.3 μs
226.7 μs
340.0 μs
453.4 μs
566.7 μs
38.7 μs (5.9×)
39.8 μs (11.4×)
64 (4^3)
512 (8^3)
1,728 (12^3)
4,096 (16^3)
8,000 (20^3)
超节点总卡数 N（对应 3D Torus 边长 K^3）